Dolby Vision Manual Calibration with DV Config Upload

The following short guide will provide details on how to manually calibrate Dolby Vision on either a 2019, 2020 or 2021 LG OLED using the internal Test Pattern Generator (iTPG), the 22 Point White Balance Controls and upload of a Dolby Vision Configuration File via the USB drive.

It must be noted that you cannot validate or verify the outcome of the calibration when using the internal pattern generator due to the location of the iTPG in the image pipeline, and the iTPG's inability to generate the required Dolby Vision metadata. But if the procedure is carried out correctly then the eventual tracking of 2.2 Gamma obtained during the manual adjustments of the 22 Point White Balance will transfer to track PQ EOTF extremely accurately when using specialised Dolby Vision test patterns containing "Absolute" metadata. Unfortunately, such patterns are only available when using very expensive reference external test pattern generators that are outside the realms of affordability for most home enthusiasts. But those that have the means to generate such patterns can verify that it works as intended, and this will be shown later in the guide due to the fact I have been able to verify using Dolby Vision Absolute test patterns using an external test pattern generator over HDM.

To begin, you must ensure you have the latest version of Device Control (currently 2.0.19) and the correct LG OLED Template for Device Control for your TV model. At the time of writing this guide, this will be LG 2019 Template Rev.5, LG 2020 Template Rev.2 and LG 2021 Template Rev.1. Everything from here will work under the assumption that Device Control is set up correctly and that you are familiar with how to operate it to connect to your TV and send the required DDC commands.

It is also advised to make your adjustments starting from a pre-calibrated Service Menu White Balance colour temperature. This can be the same pre calibrated Service Menu White Balance (SM WB) used for SDR 3D LUT calibration. There is no need to pre-calibrate separate SM WB settings just for Dolby Vision or HDR.

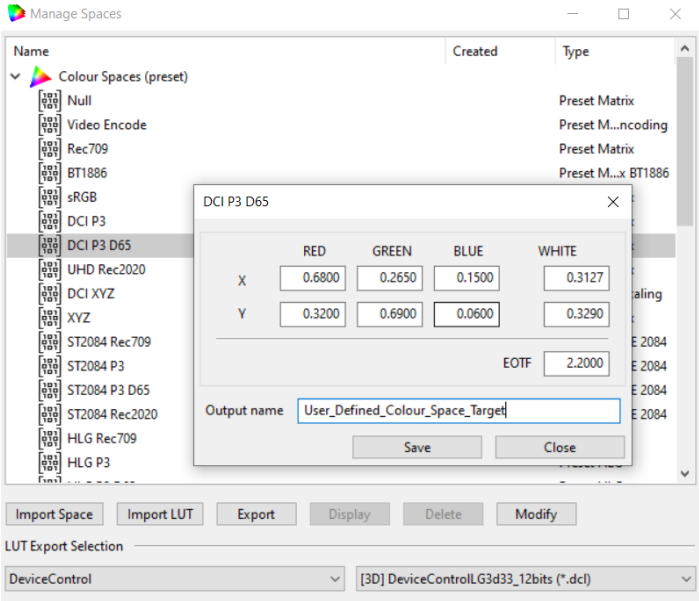
It is recommended that when calibrating Dolby Vision, it is adjusted to a reference D65 (0.3127/0.3290) white point only. Whilst it is possible to calibrate to an alternate white point, it has since been discovered that the white point shifts when receiving certain types of Dolby Vision signal, which can invalidate the calibration. This is due to the internal Dolby processing within the TV basing all of its calculations on an underlying D65 white point.

(Note: While I have previously calibrated Dolby Vision to an alternate white point myself in the past, I have since discovered a lot more information about how the processing of the signal occurs. I will not be sharing this information at any time!!!! But it is highly advised you do not attempt to adjust to an alternate white point in Dolby Vision. This advice/warning does not apply to SDR or HDR where an alternate white point can be used)

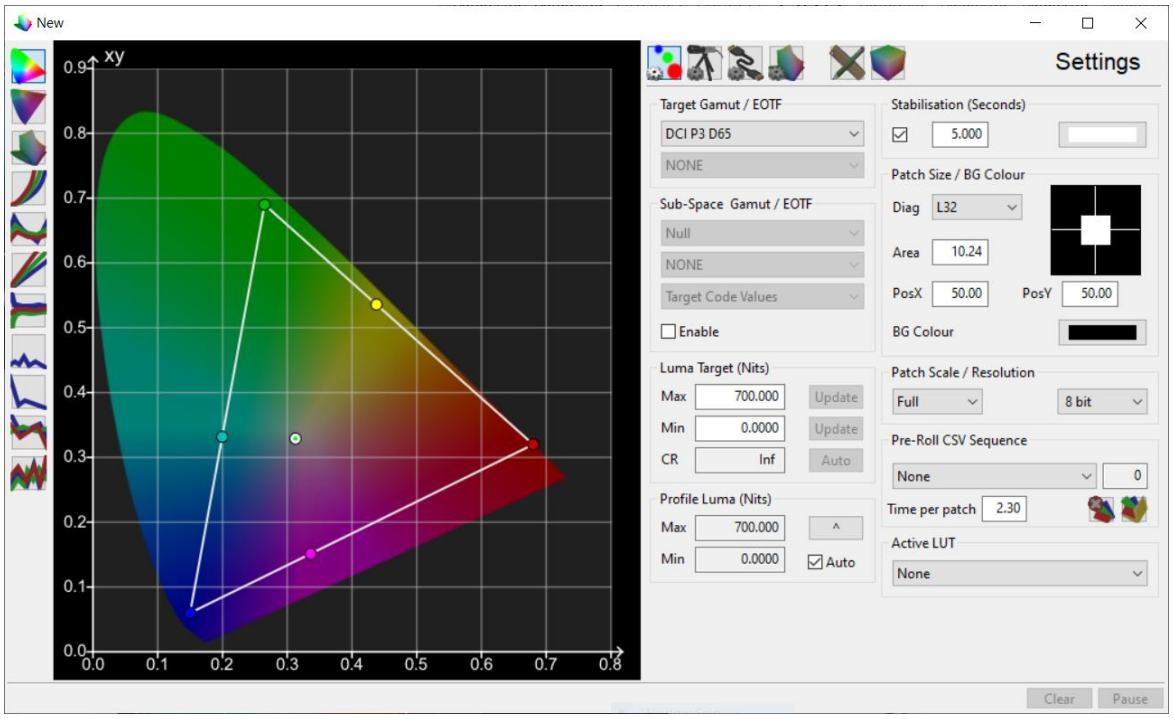
When calibrating Dolby Vision with this method, you will be adjusting your white balance controls to target a Power Law Gamma of 2.2. The conversion to PQ EOTF is done automatically based from these underlying adjustments. I am including a modified colour space target within the package you have received, but if you wish to create your own, you can do this from the “Manage Spaces” window.

Open the section for Colour Spaces (preset) by clicking on the drop-down arrow.

Select DCI P3 D65 and click Modify to edit the EOTF value to 2.2000. Give the new modified Colour Space a name and click Save.



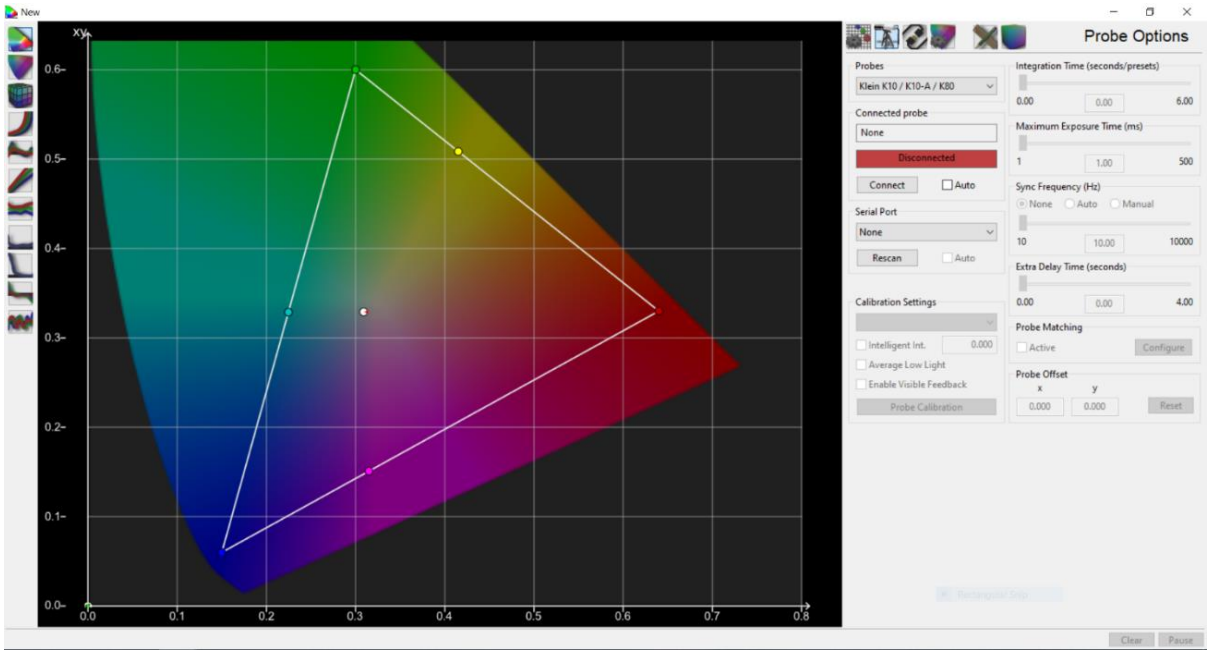
Open a ColourSpace Profile window and set your Target Gamut now.



While in this screen and after setting Target Gamut, set Stabilisation to 5 seconds and tick to enable. Set Patch Size Diag to L32. Place a tick in Profile Luma (Nits) Auto and set Patch Scale to Full.

Change to the Probe Options tab and connect your colorimeter, remembering to configure Probe Matching if required, or selecting a meter pre-set (e.g. EDR) if your meter is not profiled to a Spectro. Set all other probe options as you normally would (e.g. Sync Mode AIO etc, for i1 Display Pro 3 users)

Set a minimum Extra Delay Time (seconds) of 0.50 to minimise the risk of any sync read issues with the iTPG.

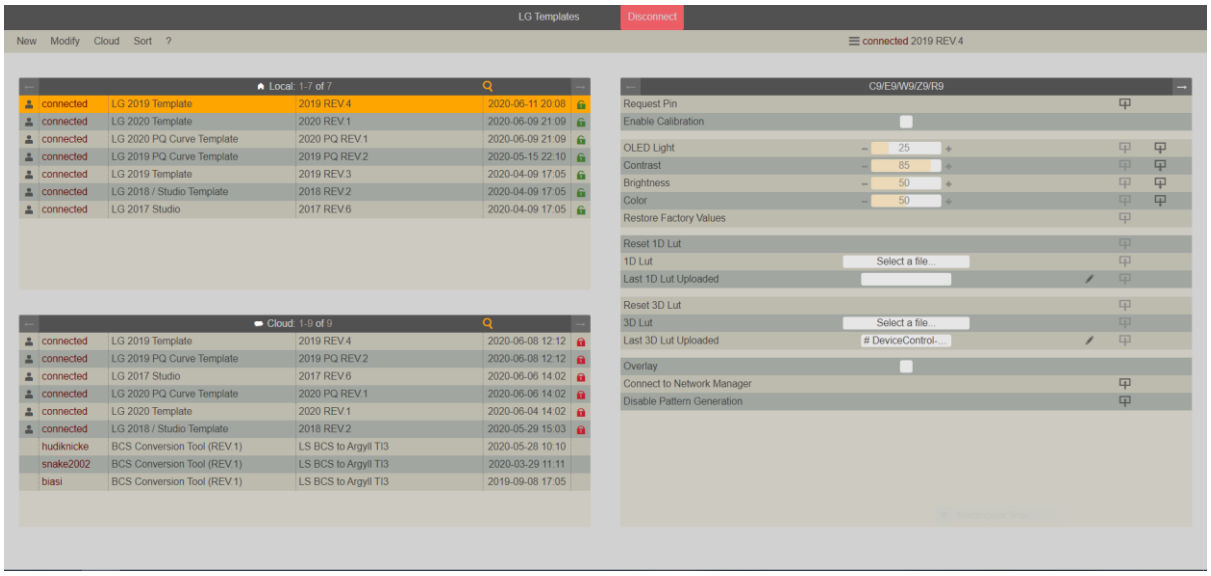


Change to the Hardware Options tab in ColourSpace in preparation to connect to the iTPG

Launch Device Control, open your web browser and navigate to http://127.0.0.1

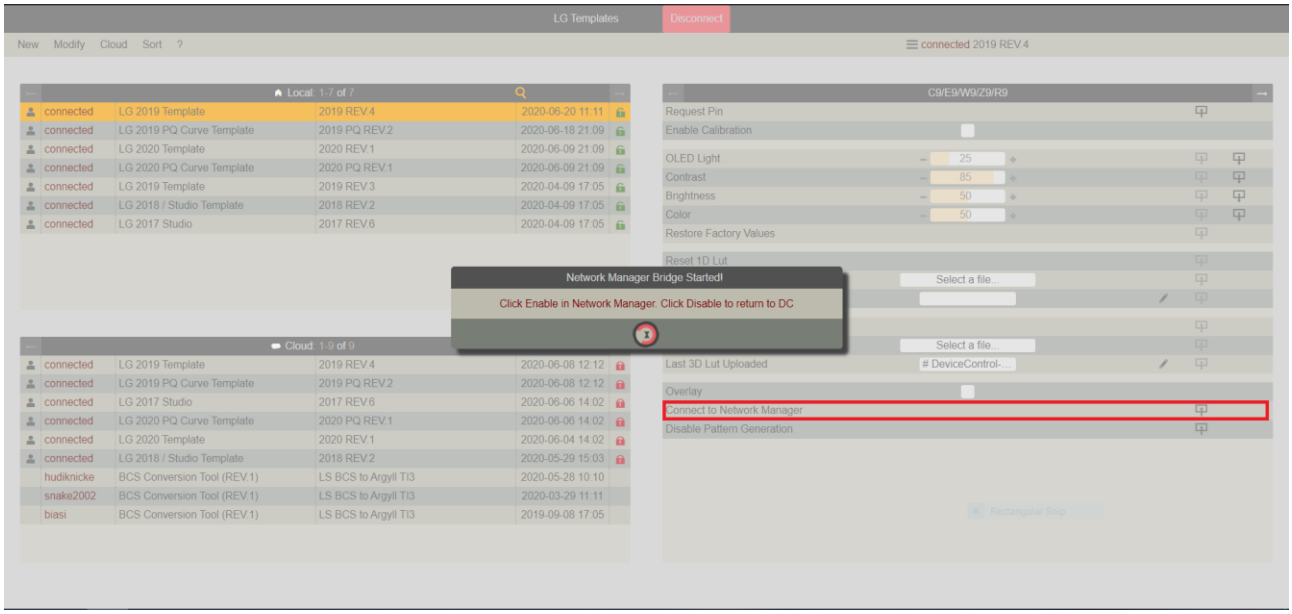
Select the LG Template for your model TV, and click the green Connect button.

Connect to your TV by Requesting PIN, then entering and sending the PIN displayed on screen.

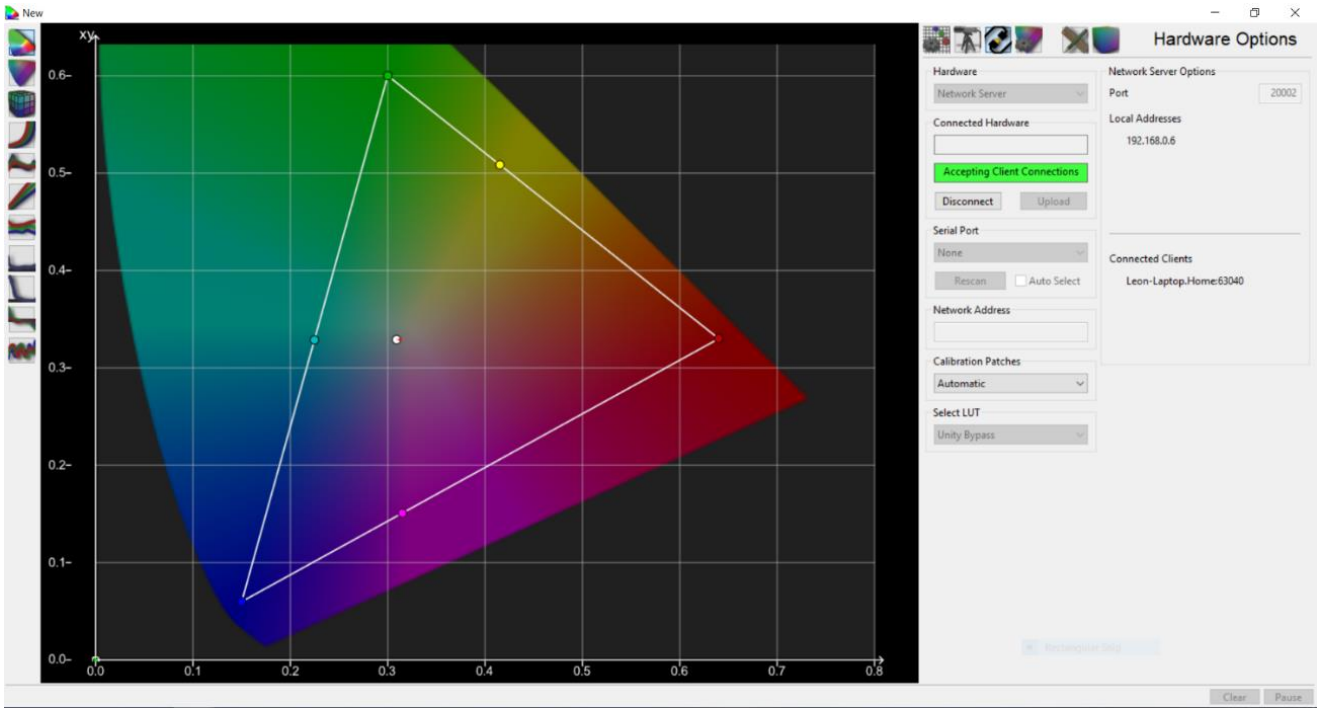


If you have a Dolby Vision blank video file to put onto a USB drive, insert this into the TV USB port and play it via the internal media player now. If you do not have a Dolby Vision blank video file, I have provided one within the package you have received. Failure to keep Dolby Vision video in the background could mean the iTPG dropping back out of Dolby Vision mode and you will end up adjusting to SDR luminance levels.

With the TV now in Dolby Vision mode, select “Connect to Network Manager” in Device Control Template, and switch (e.g. Alt+Tab) back to ColourSpace

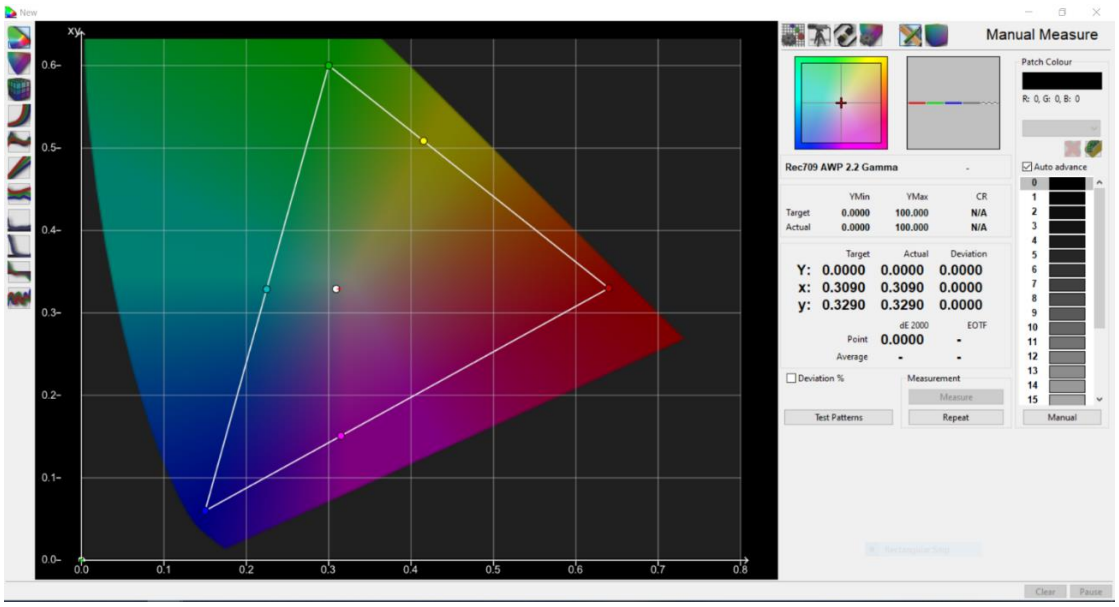


Select Network Server from the Hardware list and click Connect. You will see your computer name shown in the Connected Clients. Once it appears, select Automatic under Calibration Patches. You are now connected to the iTPG in Dolby Vision mode and have the ability to display the required patches needed for this calibration.



Make sure you are in Dolby Vison Cinema (not Cinema Home) at this point, and that your Colour Temperature is set to the same one you have pre-calibrated in the Service Menu. Please also ensure all extra video processing features are disabled, such as Noise Reduction, etc.

Change to the Manual Measure tab, click “Presets” and load the provided LG OLED 22 Point White Balance.csv patch sequence that has been included with the guides.



Place a tick in “Auto Advance” and click Measure to take a pre-calibration grey scale sweep. Once this has completed, **you must remove the tick from “Auto Advance”** before progressing to the next step.

Click Repeat, then Measure, and starting from patch number 22 (100% White), begin adjusting the 22 Point White Balance controls from the TV User Menu -> Picture Settings -> Expert Controls.

Please note that the 5 seconds stabilisation patch insertion mentioned previously in this guide is only my personal suggestion at this time *(see the added note below)* but may need adjusting depending on how much your readings fluctuate. If they remain stable then the 5 seconds value should suffice, but if you note that they fluctuate significantly, you may need to raise this value.

This also prevents possible image retention with such high luminance patches being displayed on screen for too long, and also helps when high luminance patches are displayed for too long a period on a WRGB OLED panel, as the luminance value fluctuates significantly which makes it nearly impossible to adjust as it will always be a constantly moving target.

Once 100 IRE is successfully balanced and on target, continue to work your way through the patch sequence down to patch number 12 (50% Grey). Once you reach 50% Grey you can momentarily stop measurements to change to the Settings tab and change the Stabilisation value to 2.5 seconds. At 50% Grey luminance levels, the panel is a little more stable and this will speed up the remainder of the procedure.

Change back to Manual Measure and continue to adjust from patch number 12 down to patch number 8 (30% Grey). Once you reach 30% Grey you can momentarily stop measurements to change to the Settings tab and change the Stabilisation value to 0.1 seconds. At 30% Grey the luminance is much more stable and this will speed up the remainder of the procedure significantly.

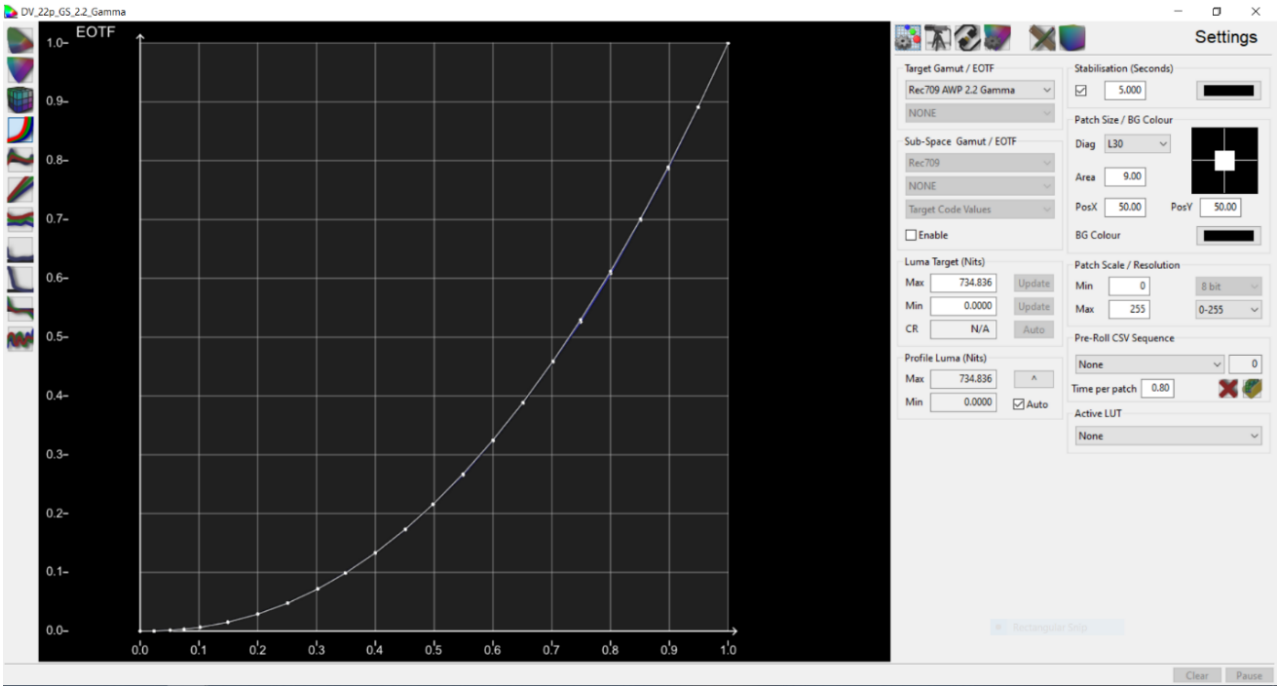
Change back to Manual Measure and continue to adjust from patch number 8 down to patch number 1 (2.5% Grey).

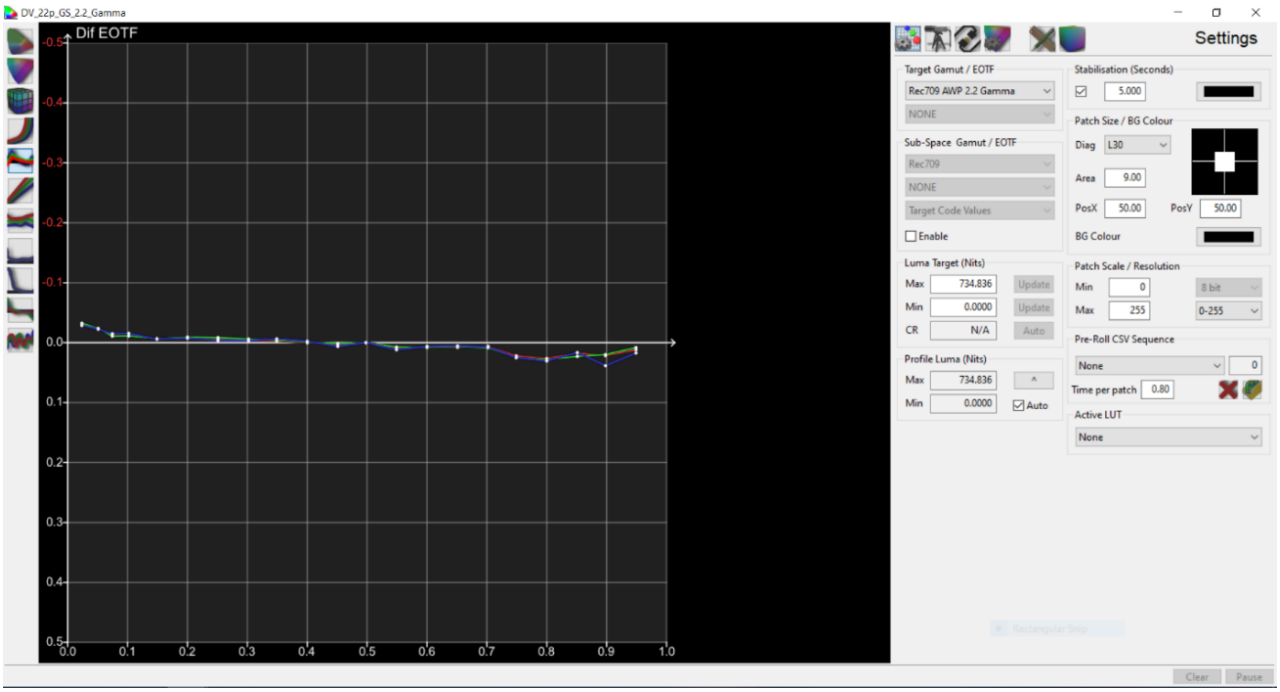
Note: When using an i1 Display Pro probe, Stabilisation length may need increasing up to as much as 10 seconds or longer for 100% White and 95% Grey to prevent luminance rising during adjustment. This is due to the slightly slower read time when compared to the Klein K10-A (used when creating this guide) causing the panel to heat and peak luminance to continuously rise.

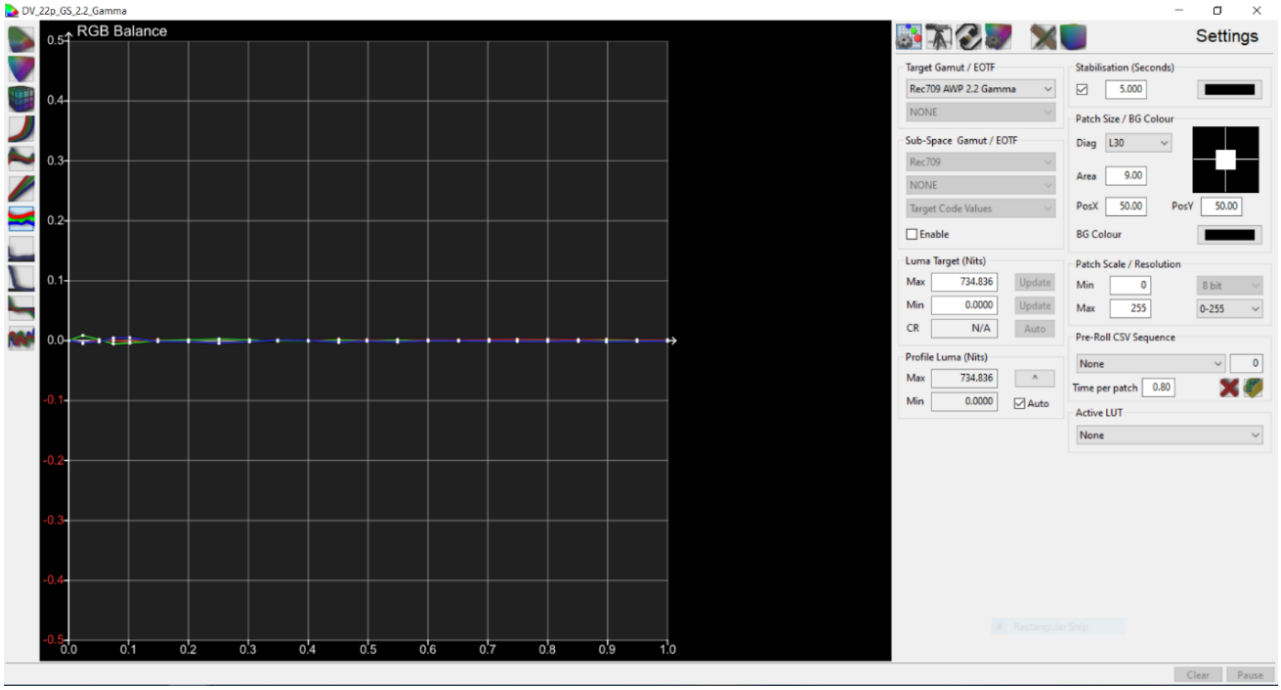
Once all 22 Points have been adjusted you can, if you like, set Stabilisation back to 5 seconds, place a tick Auto Advance and remeasure the full 22 Point sweep to confirm your adjustments are correct or whether they need some additional fine tuning. But it must be remembered that once 100% white is remeasured, if luminance has fluctuated and is different than before, then it will mean that all target points from 2.5% to 95% will also change because at this point you are measuring in gamma space, which is a relative standard.

For example, if your first sweep recorded a peak white value of 700 nits, then 95% Grey would be 682 nits. But if your second sweep records a peak white of 710 nits, then 95% Grey now has a target of 691.7 nits. This example will apply to all points in between too.

The ultimate intention is to have good 2.2 Gamma tracking along with good RGB Balance. When this is achieved, then your EOTF, Diff EOTF and RGB Balance graphs should be looking something like those shown below.





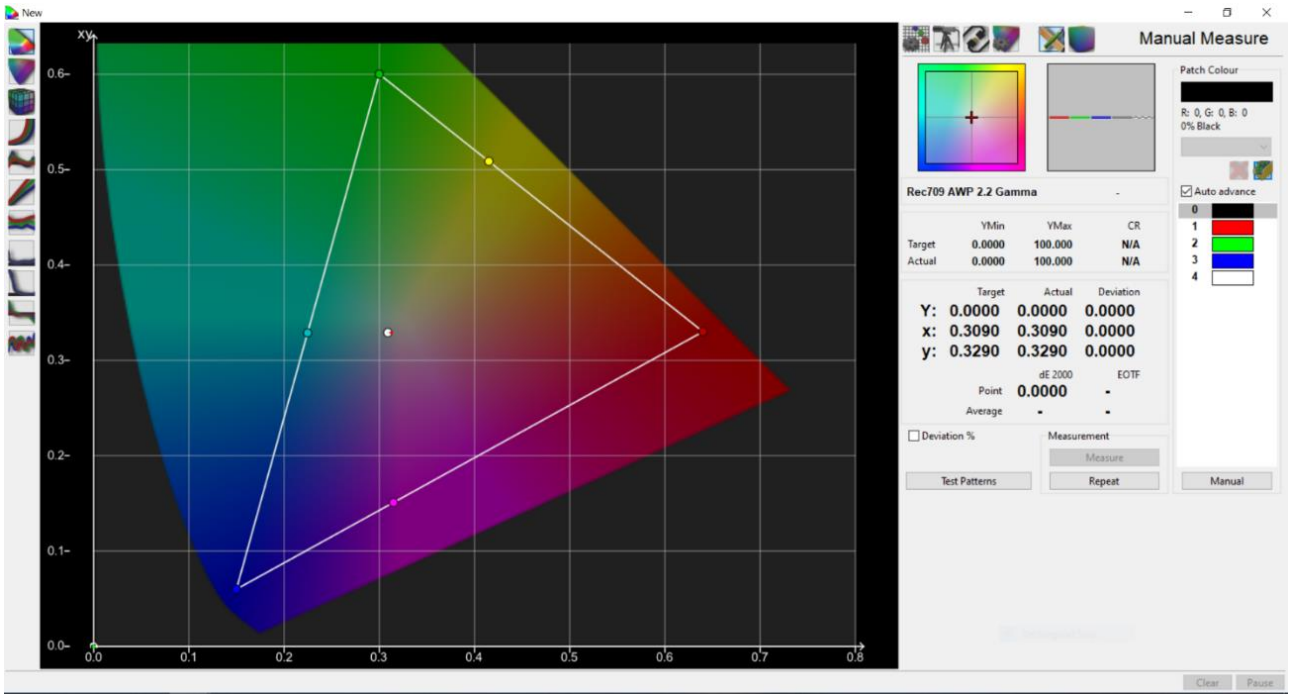


Results as shown above will provide excellent PQ EOTF tracking after the internal processing conversion (this will be shown later in the guide).

When you are happy with your adjustments, it is essential remember at this point to go back to the User Menu -> Expert Controls -> White Balance and select "Apply to All Inputs"

With the greyscale/white balance complete, it is time to measure the primary colours and create/edit a Dolby Vision Configuration file for upload to the TV.

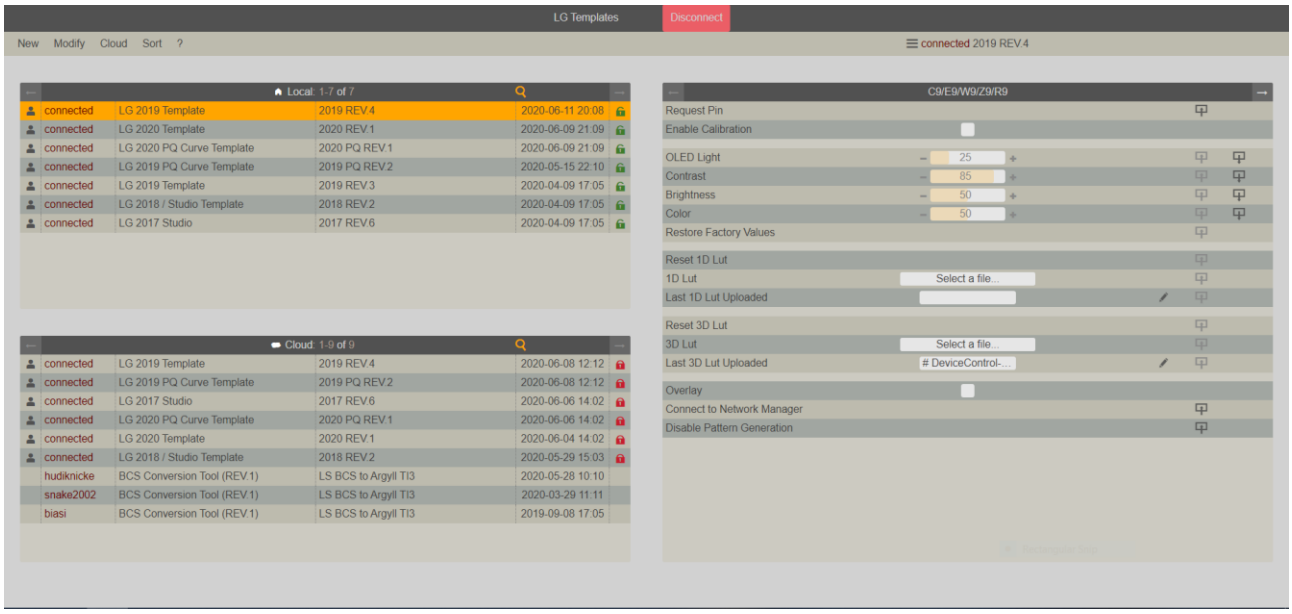
Change to the Manual Measure tab and load the Black,R,G,B,White.csv patch sequence that has been provided.



Tick Auto Advance and click Measure to record the values of each patch. Once this has completed, all required measurements have been taken so you can now disconnect from the iTPG.

Change to the Hardware Options tab and click Disconnect.

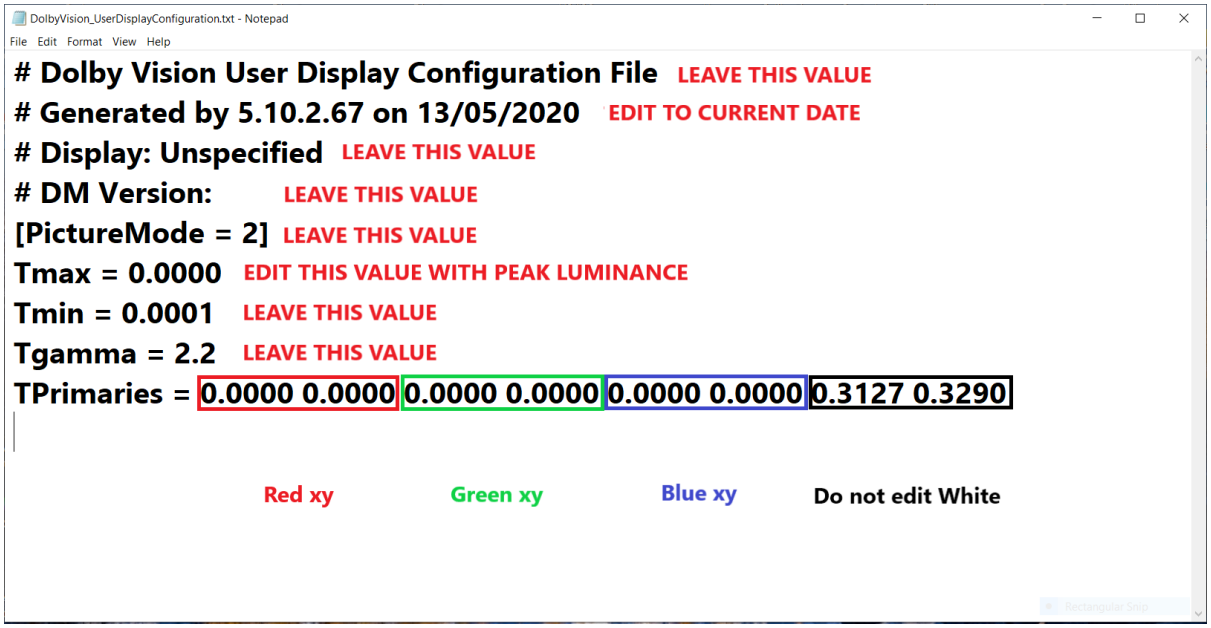
Switch (Alt+Tab) to the Web browser window and click to **“Disable Pattern Generation”** to fully disable the iTPG.



Failure to “Disable Pattern Generation” will result in the iTPG continuing to be displayed despite not being able generate patterns from ColourSpace.

You can now disconnect Device Control Template and close the Device Control command window.

Open the provided DolbyVision_UserDisplayConfiguration.txt file provided, ready for editing with the peak white value and recorded primary colour coordinates.



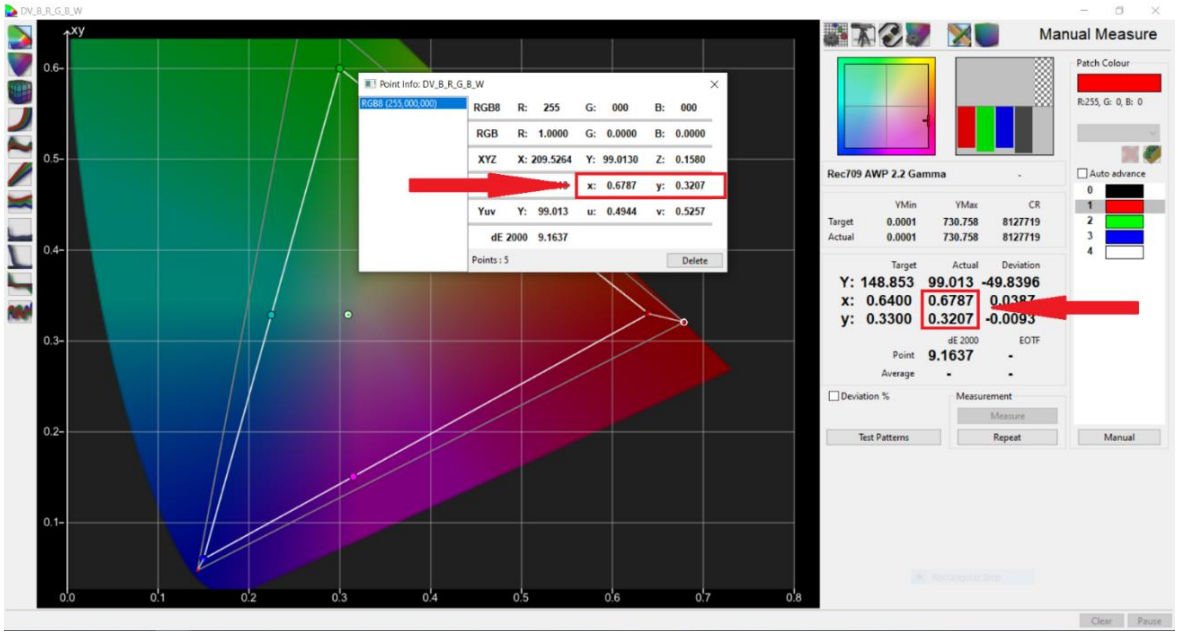
<u>As noted in the above image, some values/lines must be left as they are and some should be edited It is critical that the formatting remains intact, and only the specified values are edited.</u>

<u>It is recommended to leave the Tmin value as it is (0.0001) even if your black measurement is registering 0 Nits in ColourSpace.</u>

<u>Also, it is essential that you do not edit the White coordinates in the Configuration file, regardless of your actual measured xy coordinates of your peak white.</u>

Enter peak White luminance (Nits) into the “Tmax =” line, and the coordinates (xy measurements) of Red, Green and Blue into the “TPrimaries =” line by selecting each patch from the patch sequence window and noting the Actual measured xy data, or double clicking the individual points in the CIE chart and noting the xy data from the Point Info window as shown below.

Note: - Ignore any/all reported dE values as they have no relevance to the procedure. The purpose is to record the panels luminance and gamut capability only.



Save the edited DolbyVision_UserDisplayConfiguration.txt file, and place it onto a USB drive.

Note: - Most standard USB drives should work if formatted to either FAT32 or NTFS, but some users have had issues with some drives in the past when they have been formatted to NTFS, whereas FAT32 seems to work more consistently. Whilst just copying the file to any USB drive normally does “just work”, other files being on the drive can also at times cause issues. If this is the case, then format the Drive to FAT32 making sure there are no other files on the drive.

Insert the USB drive containing the DolbyVision_UserDisplayConfiguration.txt file into the USB port on the TV, if a pop up appears to select Photos/Video etc, just close it or wait for it to disappear. With the USB inserted, you must now play some Dolby Vision content. This can either be from an internal app or an external device. My quick and easy choice is the Dolby Access app (free from the WebOS app store) but Netflix or any other internal app that is Dolby Vision compatible will work, as will an external device such as the ATV4k. Anything will work as long as it can put the TV into Dolby Vision mode.

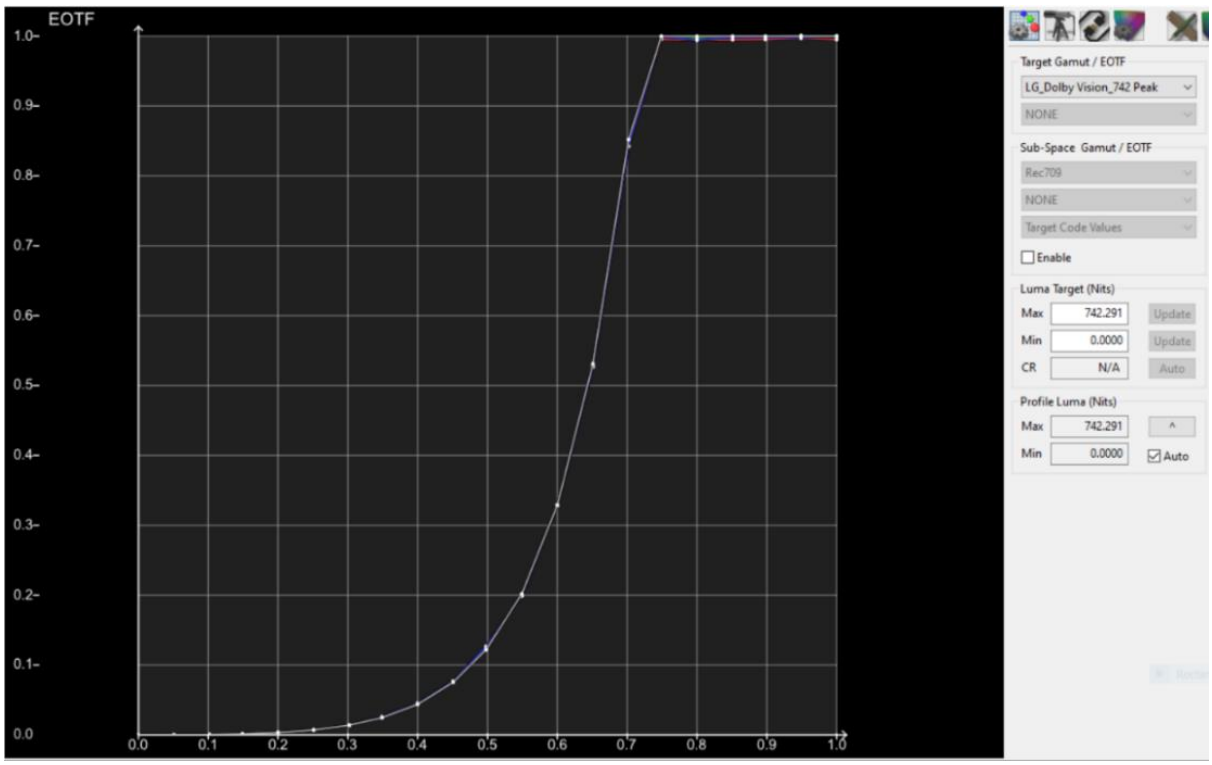
After a few moments a pop-up message should appear on screen stating a new Dolby Vision configuration has been found. Click to Update CFG.

Stop the Dolby Vision content playing to put the TV back into SDR mode, and then finally play the Dolby Vision content one more time to fully enable the uploaded CFG.

Dolby Vision is now calibrated, and the uploaded Dolby Vision configuration has been activated.

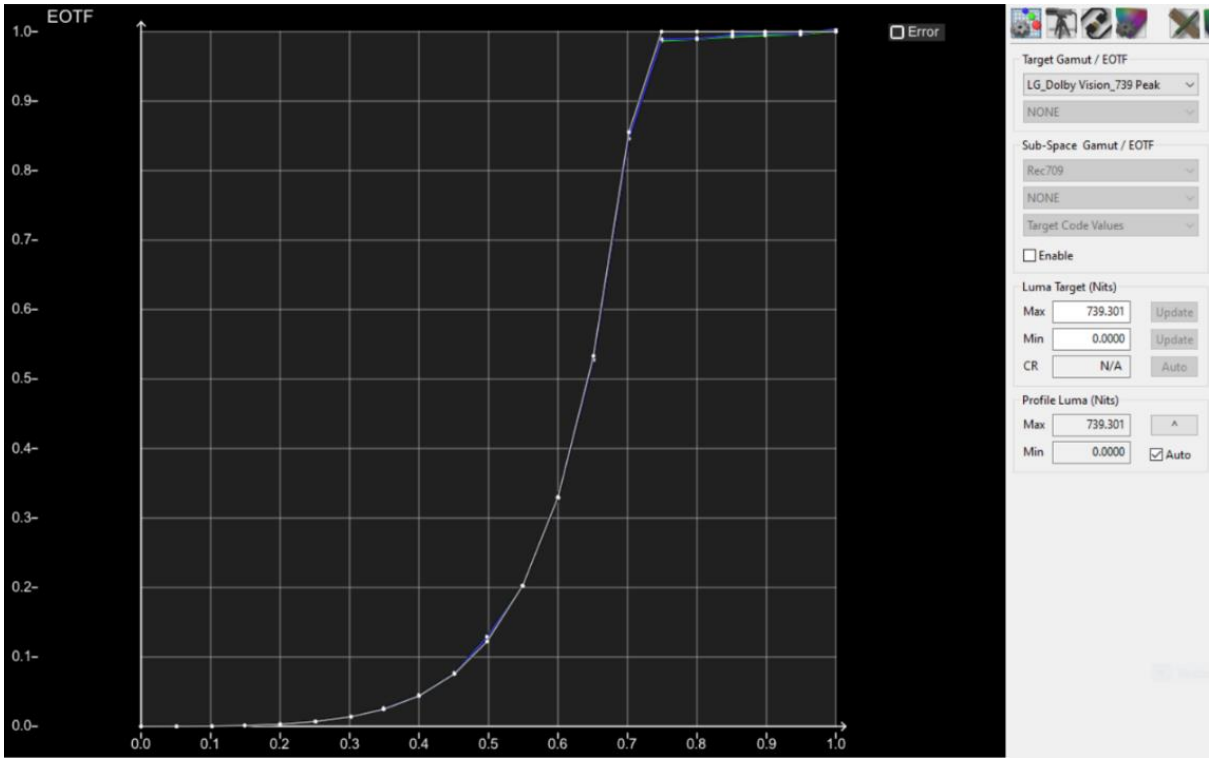
To see how the above adjusted points convert to PQ EOTF tracking, I have verified it with both an external TPG sending Dolby Vison Absolute 20 Point grey scale patterns.

PQ EOTF tracking with DV Absolute patterns over HDMI



For comparison, I have also measured Ryan Masciola's Dolby Vision Pattern Suite, 20 Point Greyscale also, which are Full Range, Profile 5 Dolby Vision patterns.

R.Masciola's Dolby Vision Test Pattern Suite – 20 Point Greyscale



With Dolby Vision in Absolute mode, PQ EOTF should be tracked up to the panel's peak luminance output capability and then it should hard clip. This is the nature of PQ EOTF as it is an absolute standard. As can be seen in the above two images PQ EOTF tracking is perfect and hard clips at the peak limits of the panel, confirming that the calibration to 2.2 Gamma has been successful, as has the internal processing conversion to PQ EOTF.

With the upload of the Dolby Vision Configuration file that contains the actual measured data of the panel's luminance and gamut capabilities, it will now enable the Dolby Vision on board processing engine to more accurately "tone map" and "gamut map" when receiving real world Dolby Vision content containing dynamic metadata.

In other words, it works very well indeed!

With the procedure understood and your reference Dolby Vision Cinema picture mode calibrated to its most accurate potential, it is now possible to also calibrate Dolby Vision Cinema Home, and Dolby Vision Game picture modes too.

Thanks to the help of AVS Forum user, **HiFi4Vision**, we have discovered that it is indeed possible to apply a Dolby Vision configuration file to these picture modes also. It just requires some extra manual editing of the DolbyVision_UserDisplayConfiguration.txt file.

But first, it is worth noting that both of these modes have their limitations that are beyond the users control to "calibrate out".

Dolby Vision Cinema Home:

By its very nature intentionally breaks accurate tone mapping to overcome brighter environmental surroundings. Luminance is boosted in the lower to mid tones to give an overall brighter average picture level. However, with the Ambient Light Sensor, Dolby Vision IQ can mean these picture modes have some benefit by having a more accurate calibration.

(Note:2019 LG OLEDs do not have a true Dolby Vision IQ mode as it was an early implementation, but the Ambient Light Sensor means it performs in much the same way as true Dolby Vision IQ found on 2020 & 2021 models)

Dolby Vision Game:

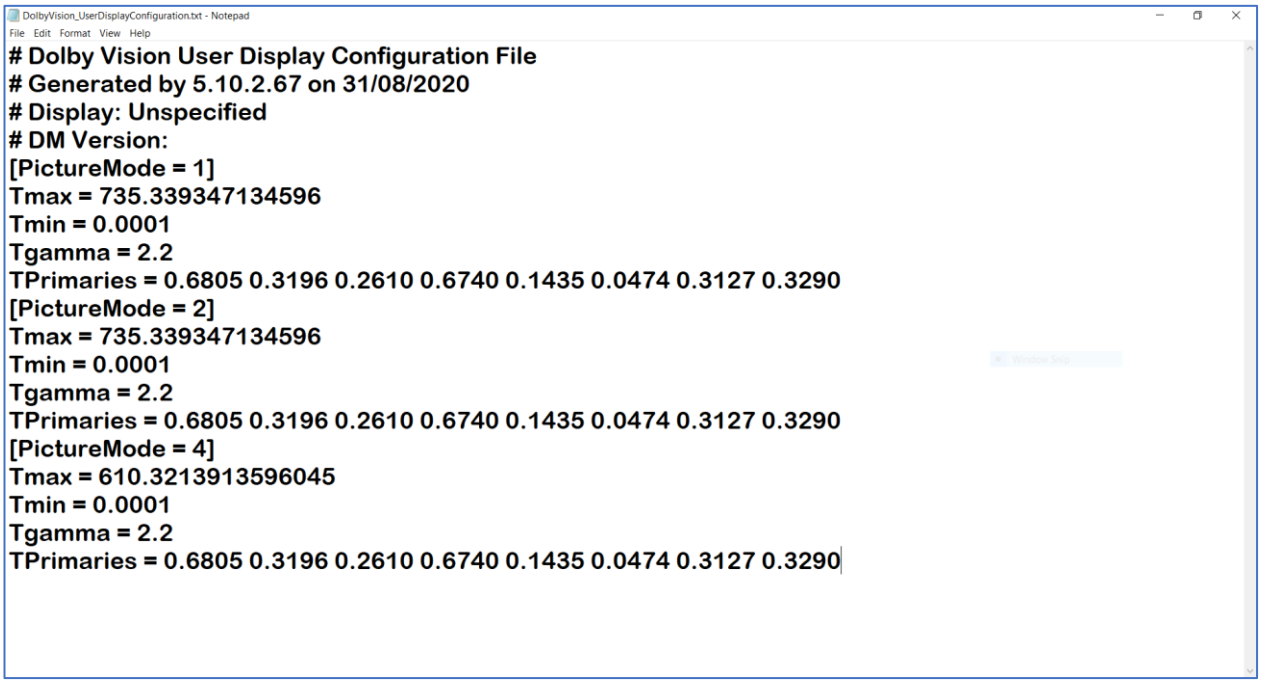
This picture mode has a substantially reduced peak luminance. The general consensus is that this picture mode is intentionally driven less to prevent the chance of burn in from bright static elements that remain on screen for a significant amount of time when gaming (video game HUDs, etc). Due to this reduced peak output, the loading of a configuration for this mode helps with more accurate tone mapping. With Dolby Vision games soon to become a thing, users may find potential in applying the information to the file for this picture mode.

It is up to you, the end user, as to whether you go to the lengths of adjusting the multipoint white balance controls. If you choose to do this, then the procedure is identical to that explained above. With the obvious exception that Dolby Vision Game is going to measure much lower at 100% peak white.

Regardless of whether you choose to adjust the multipoint white balance or not, some additional information needs to be added to the file, but it is absolutely critical that it is entered correctly with no formatting changes, or the file will be seen as "invalid" and will be bypassed *(ignored)* by the Dolby Vision processing engine.

I am providing a second **DolbyVision_UserDisplayConfiguration(EXTENDED).txt** file that has been modified to contain the extra lines required in the correct format the TV is expecting. When loading in this format, the configurations will successfully apply to the relevant picture modes.

When correctly edited, it will look like this



```
# Dolby Vision User Display Configuration File
# Generated by 5.10.2.67 on 31/08/2020
# Display: Unspecified
# DM Version:
[PictureMode = 1]
Tmax = 735.339347134596
Tmin = 0.0001
Tgamma = 2.2
TPrimaries = 0.6805 0.3196 0.2610 0.6740 0.1435 0.0474 0.3127 0.3290
[PictureMode = 2]
Tmax = 735.339347134596
Tmin = 0.0001
Tgamma = 2.2
TPrimaries = 0.6805 0.3196 0.2610 0.6740 0.1435 0.0474 0.3127 0.3290
[PictureMode = 4]
Tmax = 610.3213913596045
Tmin = 0.0001
Tgamma = 2.2
TPrimaries = 0.6805 0.3196 0.2610 0.6740 0.1435 0.0474 0.3127 0.3290
```

Whereby, the values under each [PictureMode=#] relates to a specified picture mode

[PictureMode=1] = Dolby Vision Cinema Home

[PictureMode=2] = Dolby Vision Cinema

[PictureMode=4] – Dolby Vision Game

It is critical that the format of the file is exactly as shown above, or it will be bypassed (ignored) and the configuration will not be applied correctly to the calibration.

The file I am providing is a "blank" version of the one shown in the image above for you to enter your own measured values into. When editing the file, be sure to enter a date that is later than any date you have entered into a previously uploaded file, or you may not get the update prompt pop-up.

Also, be sure to edit the name of the file **DolbyVision_UserDisplayConfiguration(EXTENDED).txt** by removing the word (EXTENDED) for it to be named **DolbyVision_UserDisplayConfiguration.txt**

Once again, credit goes to AVS Forum user **HiFi4Vision** for his assistance in finding the ability to upload the file in this manner.

Happy viewing,

Liberator72